CQE Paper 32 - The Supervisor Cursor: Superpermutations as the Compressed Dimensional Action Graph

CQE-paper-32
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Use superpermutation schedules as supervisor cursors: compressed action graphs that enumerate every requested ordering while producing deployable PermForge/GraphStax machinery.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- production paper body pending rewrite: production\papers\CQE-paper-32\PAPER-BODY.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-32\paper_kernel_manifest.json
- body: production\papers\CQE-paper-32\PAPER-BODY.md
- source: production\papers\CQE-paper-32\SOURCE.md
- formal: production\papers\CQE-paper-32\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-32\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-32\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-32.md

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Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, lemmas, constructions, examples, and receipts that establish the claimed transport. Paper 00, workbook sheets, analog tools, and open-obligation ledgers are supplemental validation and exposure layers. They exist to make the math inspectable, reproducible, and accessible without requiring a particular software stack. In the simplest case, the same state transitions can be marked with ordinary physical tokens, lines, or dirt; the point is not the material, but the preserved center, boundary, transform, residue, and receipt structure.

Series note. Papers 00-31 built the substrate. Paper 32 opens the applied series: the methods of the corpus pointed at problems that look arbitrary from outside - and the demonstration that solving them produces working machinery (here: PermForge, the supervisor-cursor engine of ChromaBlend Studio) while the problem's own structure lands, term for term, inside the corpus mathematics.

1. The seemingly arbitrary application

The superpermutation problem asks: what is the shortest string over n symbols containing every permutation of those symbols as a contiguous substring? It reads as a curiosity of recreational combinatorics. It is not. Within this corpus it is the exact statement of a question the substrate already forces:

If C only exists when an enumeration is requested (Gamma(s) = _C(enum(r_i, W)), Paper 15's emission law read operationally), then what is the complete, minimal schedule of all enumeration requests at scale n?

That schedule is the superpermutation. The system reads dimensions via tape in slots; an ordering of n slot-reads is a permutation; the set of all orderings is the dimensional action graph at D = n (n! vertices); and the superpermutation is that graph serialized onto a 1D tape with maximal overlap sharing - its **full compression**:

> superperm(N) = compress(ActionGraph(D = N))

Live numbers: 24 readings in 33 symbols (n=4, 2.91× over naive), 120 in 153 (n=5, 3.92×), 40320 in 46205 (n=8, 6.98×). The cursor string is a supervisor, never content: walking it fires every possible ordering of slot-reads exactly once.

2. Walking the power-of-ten ladder: solving each N+1

The classic recursive construction is the chart walk made literal. To lift scale n to n+1: visit every permutation of scale n **in the order the cursor visits them**, and thread the new symbol through each one (block p·(n+1)·p, merged with maximal overlap). Every local chart of scale n becomes a sheet of the (n+1)-cursor - local at resolution R is global at resolution R-1, the MDHG ladder walked one rung per symbol.

Executed live from the single character "1" (total walk time 0.04 s):

| n | length | log10 | = k! | coverage | MDHG rung |

|---|--------|-------|-------|----------|-----------| | 1 | 1 | 0.00 | yes | yes | grain | | 2 | 3 | 0.48 | yes | yes | dust | | 3 | 9 | 0.95 | yes | yes | triad | | 4 | 33 | 1.52 | yes | yes | block | | 5 | 153 | 2.18 | yes | yes | cluster | | 6 | 873 | 2.94 | yes | yes | domain | | 7 | 5913 | 3.77 | yes | yes | region | | 8 | 46233 | 4.67 | yes | yes | planet |

Each N+1 solve is one rung up the power-of-ten scale - log10 advances by ~= log10(n) per rung, the factorial walk. The ninth rung (universe) is the asymptote the ladder names but does not reach: there is no terminal N.

3. All complexity effects are inside the local charts

Everything difficult about *minimality* is a property of the length-n windows - the local charts - not of the global string:

- **n=4 (block):** the chart structure admits exactly ONE minimal cursor, and it is palindromic - equal to its own LR-podal reversal. 24 permutations = the 24 roots of D4 = the vertices of the 24-cell, the unique self-dual regular 4-polytope. A 4D object with a self-mirrored schedule: no torsion.
- **n=5 (cluster):** minimality settled by exhaustive search (Chaffin, 2014): exactly **8 minimal cursors** of length 153 - one palindrome and seven symmetry-broken trees. **8 solutions = 8 dimensions = the E8 lane count.** The 4D object's schedule, lifted one symbol, demands the full 8D ambient space. The seven trees admit no canonical choice: under relabeling × reversal the octad is a homogeneous space - a torsor. Choosing a cursor is choosing a gauge. This is the GR torsor effect of the dimensional lift.
- **The wrap correspondence (recorded observation, pending validation):** the reversal involution acts on the octad with 4 fixed points + 2 swapped pairs ({0,1,4,6} fixed; 2<->5, 3<->7) - the same orbit type as the 8 chart states under swap_LR (4 Lie conjugates fixed + 2 chiral pairs), the involution whose invariant defines the gluon Gamma(s) = C. The schedule space at n=5 and the state space at n=3 wrap identically.
- **n>=6 (domain and up):** the charts stop admitting closed answers. Minimality is open; only bounds and search survive. The chart walk gives k!; SAT search broke it at n=6 (872 < 873, Houston 2014); the question becomes a corridor between a proven floor and a constructed ceiling.

4. The n=8 attempt - standing on Egan, Houston, Johnston, Chaffin

Nothing here is reinvented. The lower bound is the Houston-Pantone-Vatter formalization (2018) of the 2011 anonymous bound:

> L(n) >= n! + (n-1)! + (n-2)! + n - 3

The ceiling is Egan's construction (2018, building directly on Houston's minimality break):

> L(n) <= n! + (n-1)! + (n-2)! + (n-3)! + n - 3

Their difference is exact and structural: **Egan - floor = (n-3)!** - the open corridor at every scale is one factorial wide, three rungs down the ladder.

The n=8 attempt, executed live (0.06 s):

- **Chart-walk construction from scratch:** 46233 symbols (= k!), full coverage of all 40320 permutations of 8 symbols - verified.
- **The shipped field record** (curated per Johnston's records): 46205 symbols, full coverage - verified.

- **46205 = Egan's formula at n=8 exactly** (40320 + 5040 + 720 + 120 + 5). The best known n=8 cursor IS Egan's construction; it saves precisely 28 symbols over the chart walk.
- **The open window:** floor 46085, ceiling 46205, width **120 = 5! = (n-3)!**. Search has entered this corridor at smaller scales (n=6: -1 below the formula; n=7: -2, Egan 2019); at n=8 the corridor is untouched.

This is the best solution for n=8 the established methods admit, confirmed inside this format: their constructions and bounds, our verifiers and charts. The (n-3)! corridor is the standing obligation.

5. What the application produced

Solving this "arbitrary" problem inside the corpus produced operational machinery, not commentary:

- **PermForge** (`GraphStax.permforge`): embedded n=4/n=5 cursor tables, the octad with its computed reversal orbit, the bounds ladder, the N->N+1 lift, coverage verifiers, and the live n=8 exhibit - all import-time lookup, no recomputation.
- **The supervisor scheduler** (`SuperPermScheduler`): n=4 blocks of services dispatched in cursor order - every ordering of 4 services fired exactly once per 33-step pass (the service-wiring pattern of the donor stack, now a library primitive).
- **Supervised enumeration** (`GraphStaxEngine.enumerate_ribbon`): a ribbon's bits resolved into C-gluon Stax identities **in cursor order** - the first request per position is the C-production, repeats are idempotent lookups (f(f(x)) = f(x)). The C-term's dependence on requested enumeration is now executable, not aspirational.

6. Open obligations (Paper 32 ledger)

- **O-32.1** - The n=8 corridor: 120 symbols between 46085 and 46205. Egan's n=7 search methods (kernel graphs over 2-cycle/3-cycle quotients) transported to n=8 inside this format.
- **O-32.2** - Validate the octad-orbit <-> chart-state-orbit correspondence (4+2+2 = 4+2+2) as a theorem rather than an observation; identify the functor between schedule space at n and state space at n-2.
- **O-32.3** - Ship the search records below the formula at n=6 (872) and n=7 (5906) as field data alongside the construction records, with coverage receipts.
- **O-32.4** - The ninth rung: formalize the universe-level asymptote of the ladder (lim log10 behavior) in the corpus's scale language.

Citation chain

- Chaffin, B. (2014): exhaustive minimality at n=5 - the octad.
- Houston, R. (2014): SAT discovery 872 < 873 at n=6 - the chart walk is not minimal.
- Houston, R., Pantone, J., Vatter, V. (2018): formalization of the lower bound n! + (n-1)! + (n-2)! + n - 3 (anonymous origin, 2011).
- Egan, G. (2018-2019): the upper construction n! + (n-1)! + (n-2)! + (n-3)! + n - 3; search records at n=7.
- Johnston, N.: curation of minimal superpermutation records and data.
- This corpus: Paper 15 (emission law / observer chain), Paper 03 (S3 wraps), Paper 05 (oloid

carrier), MDHG ladder, ribbon 8-slot transport.

Compiled from CQECMPLX-Production master corpus. Verifiers executed live 2026-06-09; all components verified complete.